XRPL Validator — Daily Engineering Report · 2026-08-13

Period: 2026-08-13 00:00 to 24:00 US Eastern (2026-08-13 04:00 to 2026-08-14 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-ledger): stamount_signed_add ROUNDS its operands, not truncates** (`6798100`, xrpl-validator) An STAmount operand is already canonical in rippled — it was rounded to 16 digits when it was produced — so reducing an over-precise operand by truncation drops exactly the digit that decides the result. `norm16` truncates.
- **fix(xrpl-ledger): an INPUT-limited fill rounds the offer's output DOWN** (`ca17418`, xrpl-validator) BookStep.cpp:653, verbatim, with rippled's own rationale above it: ofrAmt = offer.limitIn(ofrAmt, inLmt, / *roundUp* / false);
- **fix(xrpl-ledger): measure a hop's carry at full precision, not by differencing** (`44c20d9`, xrpl-validator) The forward pass measured what a hop produced by differencing the sender's BALANCE. A balance holds 16 significant digits TOTAL, so whatever the sender already holds of the intermediate eats the low end of the addend — the hop is fine, the measurement is not.
- **fix(xrpl-ledger): reprice a rested remainder in BOTH directions** (`afb133f`, xrpl-validator) The one-way clamp on the reprice — take `derived` only when it moves the residual DOWN — was a workaround for #105930662 40FB322EC16C from when the reprice still used the RAW ratio. `070ca31` fixed the rate itself and that ledger is byte-exact now, so the guard had outlived its cause and did nothing except suppress the reprice wherever it needed to move the residual UP.

Independence program

No independence record for UTC day 2026-08-13; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Oracle-Bound Batch Execution Guard** (payments) Add a tfOracleGuard flag to the outer Batch (XLS-56) transaction plus two SFields, sfGuardOracleID (a PriceOracle keylet) and sfGuardPriceBounds (min/
- **Credential-Based Trust Line Auto-Authorization** (compliance) Add a bounded SField sfAutoAuthCredentials (STVector256 of up to 8 {CredentialIssuer, CredentialType} pairs) to AccountRoot, settable via a new Accoun
- **Native Fee-on-Transfer for MPT Issuances** (tokens) Add sfMPTTransferFee (basis points, 0–50000, mirroring NFTokenTransferFee semantics) and sfMPTFeeDestination (optional AccountID, defaults to issuer)

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-13 07:40	—	xrpl-validator	`6798100`	fix(xrpl-ledger): stamount_signed_add ROUNDS its operands, not truncates
08-13 16:05	—	xrpl-validator	`ca17418`	fix(xrpl-ledger): an INPUT-limited fill rounds the offer's output DOWN
08-13 17:37	—	xrpl-validator	`44c20d9`	fix(xrpl-ledger): measure a hop's carry at full precision, not by differencing
08-13 19:33	—	xrpl-validator	`afb133f`	fix(xrpl-ledger): reprice a rested remainder in BOTH directions

Times in UTC. A commit appears on the day it was committed to main or merged into it.